

ALEJANDRO BARAONA

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U.S. Citizen | Eligible for Security Clearance

EDUCATION

University of Texas at Austin

M.S., Electrical and Computer Engineering (Software Engineering) | GPA: 3.67

Austin, TX

Jan 2026 - Dec 2027

Northeastern University

B.S., Electrical and Computer Engineering | Worked 20hrs/week for Center for STEM Education for 2.5 years

Boston, MA

Sept 2023 - Dec 2025

Relevant Coursework: Scalable Machine Learning, Analysis of Random Phenomena (probability & stochastic processes), Fundamentals of Linear Systems, Electromagnetics, Distributed Systems, Systems Programming, Engineering Algorithms, Wireless Sensor Networks, Differential Equations & Linear Algebra

RESEARCH EXPERIENCE

Graduate Research Assistant - Wang Lab, UT Austin

Jan 2026 - Present

Articulatory and Speaker Adaptation for Disordered Speech Recognition

- Designed a **hierarchical speaker-adaptation architecture** for disordered-speech ASR: **Whisper** fine-tuned on a mixed-population corpus with **condition-specific LoRA adapters** on top, across **674 speakers** and 5 clinical populations on **TACC HPC clusters**.
- Cut overall WER from **32.93% to 26.78% (-6.15 pts, 18.7% relative)**, beating baseline on **all 5 conditions**; best case **39.66% to 30.50% (-9.16 pts)** on cerebral palsy.
- Ran a controlled comparison of conditioning signals across **674 speakers**: pooled training beat diagnosis-conditioned models on **all 5 conditions** (median stacking gain **0.16 pts**), and diagnosis proved unrecoverable from audio (**12.6%** stroke recall); evidence that redirected the project toward articulatory conditioning.
- Designing **articulatory-conditioned dysarthric ASR**: learning an impairment profile (tongue kinematics, lip closure, gesture timing) from paired **MagTrack**-audio data, then estimating it from audio alone at inference; removing sensors and the **per-user enrollment data** speaker-adaptive methods require.
- Collected clinical-grade acoustic and articulatory recordings used in **submitted publications**, including **MagTrack** electromagnetic articulography with an electrolaryngeal speaker; standardized archives into ML-ready datasets.

TECHNICAL PROJECTS

AleBot - Local-First LLM Client with Agentic Tool Use

Python, JavaScript

- Built an offline LLM research assistant (JS frontend + **zero-dependency Python backend**, including a **stdlib-only PDF text extractor**) with token streaming and a **three-tier memory scope** driven by LLM fact extraction, on local **Ollama** models with **no API keys**.
- Implemented a **multi-round agentic tool-calling loop**; literature search fans out to **arXiv, Crossref, and Europe PMC**, de-duplicating by DOI and title.
- Hardened endpoints against **SSRF and cross-origin abuse**: private/loopback IP rejection, Origin allow-listing, and click-gated sandboxed code execution.

VitalLink - Wearable Triage System

Embedded, PCB, BLE

- Designed **custom PCB hardware and firmware** for a BLE wristband streaming **4 vital-sign channels** (HR, SpO2, temperature, motion) to a base station computing real-time triage severity scores.
- Built a clinician-facing dashboard for trend visualization and alerting; **presented at Northeastern's 41st ECE Capstone Design Competition**.

WORK EXPERIENCE

Vigilant Biosciences, Inc. - Electronic & Software Integration Technician

Jun 2021 - Sept 2023

- Engineered an Arduino-based **5-parameter environmental sensor system** (temperature, pressure, humidity, CO2, UV) to verify device performance under **FDA compliance requirements**.
- Validated an **oral cancer detection algorithm** end-to-end and built Python UI components, cutting manual verification steps for regulatory submission.
- Characterized components with bench instrumentation and statistical analysis to confirm spec accuracy.

TECHNICAL SKILLS

Machine Learning / AI: PyTorch, Hugging Face Transformers, Whisper, LoRA / PEFT, transfer learning, fine-tuning, WER evaluation, encoder probing, LLM tool-calling, prompt engineering, Ollama, NumPy, Pandas, Matplotlib

Languages: Python, C++, JavaScript, Java, SQL, RISC-V Assembly, Bash, HTML/CSS

HPC & Infrastructure: TACC HPC clusters, multi-GPU training, Docker, Linux/Ubuntu, Git/GitHub, REST APIs, concurrent HTTP services

Signals & Hardware: Speech/audio signal processing, electromagnetic articulography (MagTrack), clinical data collection & dataset curation, sensor integration, PCB design, BLE, embedded firmware, Arduino, MATLAB, Vivado, LTSpice

Security & Systems: SSRF mitigation, CORS / origin validation, sandboxed execution, input validation

HONORS & AWARDS

National Merit Scholar | **Northeastern National Recognition Program Scholarship (\$40,000)** | 2023 HSF National Scholar | UPLIFT Program Scholar | NUAccelerate Scholarship | National Grid CECA Scholarship (\$2,000)